

DELPHI

DELaaware Predictive Hourly Insight The DelDOT Oracle

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powered by 



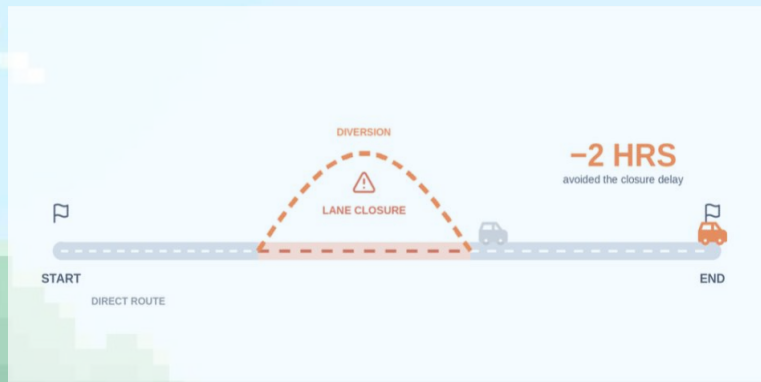
Date: 8/19/2026

**For Hackathon Presentation Use Only.
Use of the DelDOT name and logo does not imply DelDOT
endorsement of the solution.*

THE PROBLEM

Road-work planning still runs on memory **vibes**.

Every time a lane closes, someone bets on when it'll hurt least.
Nobody has a number for **this station** at **that hour** next month let alone a confidence range you can defend at a town-hall meeting.



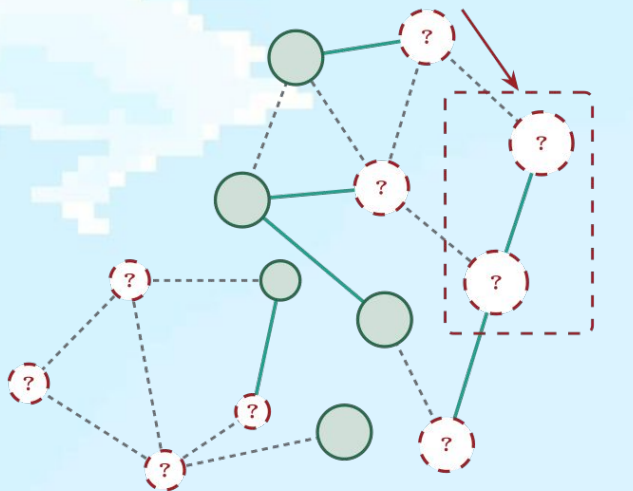
THE SOLUTION

A forecast you can **defend line by line**.
A chat layer that verifies numbers.



WHAT WE BUILT

Interstate, 100k+ veh/day on paper.
Never counted — nor was its only
corridor link, the station beside it.



recorded never switched on

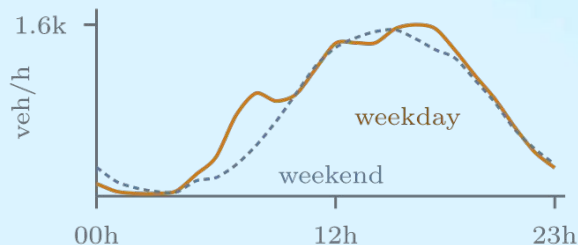
same corridor nearby

circle size = how big the road is

What we are asked for:

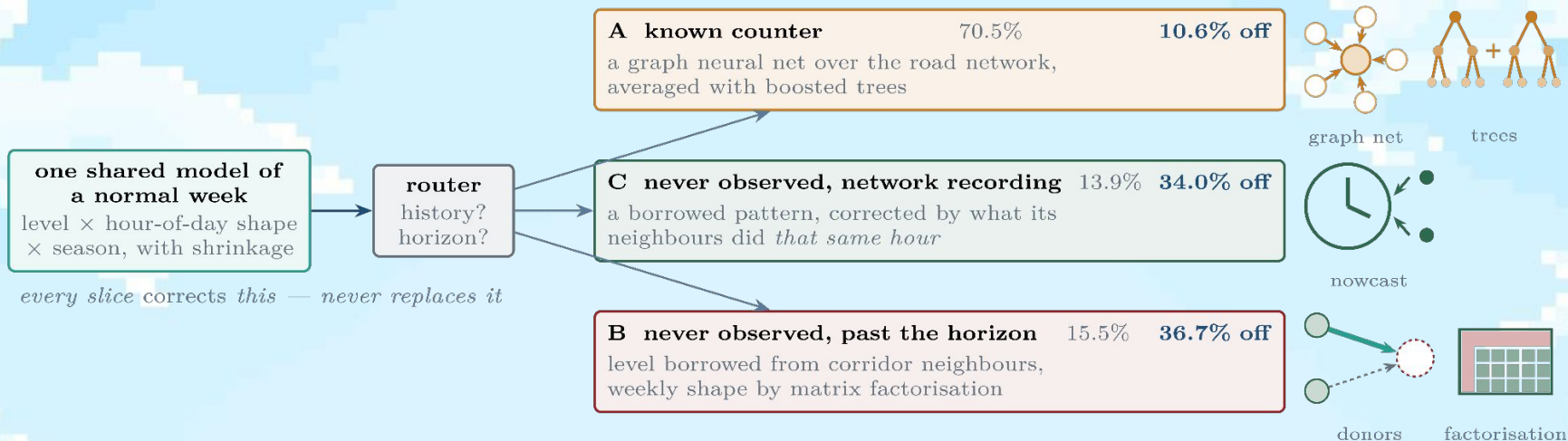
118 stations, hourly, including six months nobody
has lived through, from counts covering 75.

- Known counter, next summer. 7 rows in 10.
- Never observed, while the network was recording. No past of its own; live neighbours.
- Never observed, after the horizon. Neither.



One counter, real data. The busiest pair carries 56x the quietest.

WHAT WE BUILT



Four numbers per row, not one

Forecast. The middle of three quantiles.
low / high. The outer two, widened by a conformal offset measured per slice and volume band.
score. How much to trust the row.

The score is its own model!

A classifier predicts **P[row lands within 30%]** from slice, hour, season, interval width and history, **calibrated**, so 0.7 means 0.7.
AUC 0.851 · coverage 89.8% against a 90% target.

IMPACT

Turns every closure decision into a **defensible** number.



RESPONSIBLE AI

The LLM never writes a number.
We **enforce** it on every response.

TAKEAWAY

We didn't just call an API.
We built the **brakes** that let you trust it.



01

Problem

Planning ran
on vibes.



02

Solution

Calibrated
forecast plus
guarded
chat.



03

Result

13.4% error,
honest
intervals, live
today



04

Impact

Every closure
decision
becomes a
defensible
number